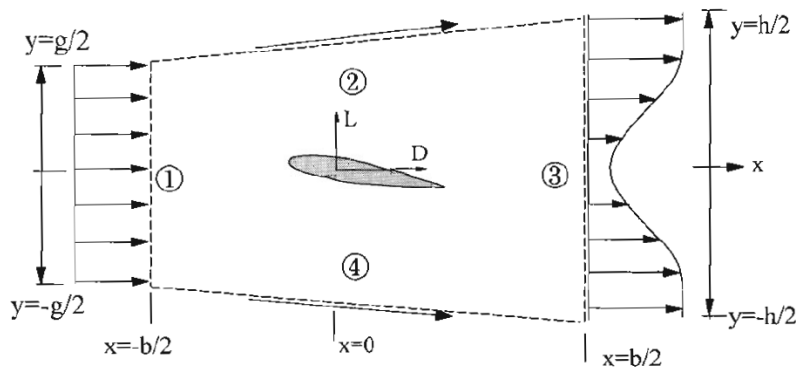


Due: 09/06/2024

1- The flow passes over an airfoil with a speed of U_∞ . The dashed line indicates the control volume around the airfoil. The upper and lower boundaries align with the streamlines, while the other two boundaries are positioned vertically. The velocity at boundaries 1 and 3 is only in the x-direction. The pressure and density at all boundaries of the control volume are constant and equal to P_∞ and ρ_∞ , respectively. The velocity at inlet 1 is uniform and equal to U_∞ , and the velocity profile on boundary 3 is shown in the figure below. U_e represents the maximum velocity. The ratio of h/g is equal to 2.292.

- Find the maximum speed, U_e , at the exit surface as a function of U_∞ .
- Calculate the drag force D as a function of ρ_∞ , U_∞ , and g
- Calculate the lift force L .

**Boundaries:**

1. Uniform velocity, U_∞
pressure P_∞
2. Pressure, P_∞
3. Velocity, $U_3(y)$ given,
pressure P_∞
4. Pressure, P_∞
5. Note: $h/g = 2.292$

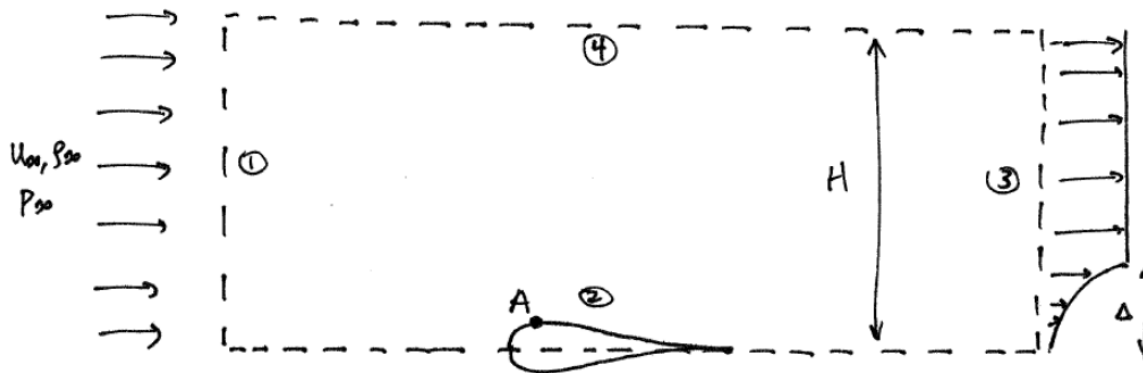
On surface 3:

$$U_3(y) = U_e \left(1 - \frac{1}{5} \cos\left(\frac{\pi y}{h}\right) \right) \quad \text{on } -h/2 \leq y \leq h/2$$

2- The velocity profile of the flow going over the surface of an airfoil in region 3 is measured as follows: ($\Delta = 0.04m$).

$$u_x = U_\infty \left(\frac{y}{\Delta} \right)^{\frac{1}{7}} \text{ for } y < \Delta$$

$$u_x = U_\infty \text{ for } y > \Delta$$



- Calculate the mass flow rate passing through surface 4 using the continuity equation.
- Calculate the drag force on the airfoil using the momentum and continuity equations. (Note that surface 2 (the lower boundary of the control volume) is a line of symmetry)